

FORMATTING INSTRUCTIONS FOR ICLR 2027

CONFERENCE SUBMISSIONS

Anonymous authors

Paper under double-blind review

ABSTRACT

1 INTRODUCTION

Large language models (LLMs) are increasingly deployed as interacting agents that exchange information, negotiate decisions, and coordinate their actions over multiple turns. Such multi-agent systems (MAS) promise to combine information and capabilities distributed across multiple agents, yet recent work has exposed substantial failures that arise specifically through interaction. Agents fail to proactively acquire information that is available only to others (Li et al., 2026a), communicate actively without converting those exchanges into effective distributed reasoning (Zhang et al., 2026b), and remain unable to coordinate effectively even when relevant information is made transparent (Yao et al., 2026). Broader analyses similarly identify inter-agent failures as a distinct source of error beyond mistakes made by individual agents in isolation (Cemri et al., 2025; Anthropic, 2026). These findings raise a basic question: what capabilities are missing when individually capable agents fail to interact effectively?

Classical models of interactive decision making provide a principled way to make this question more precise. In an interactive POMDP, an agent maintains uncertainty not only over the physical environment, but also over models of other agents, and plans its own actions with respect to this interactive state (Gmytrasiewicz & Doshi, 2005). Related work on opponent modeling similarly couples inference about another agent with planning a response to the inferred model (Huang et al., 2024). Viewed through this lens, effective interaction requires at least two computations: **interactive-state inference**, which construct belief of other agents relevant information, and **interaction planning**, which uses that belief to evaluate what to do next. Importantly, these computations are coupled over time. An inferred state changes the value of an interaction action; the selected interaction then changes how other agents respond and therefore what evidence becomes available for subsequent inference. Established models of multi-agent decision making identify these two computations as fundamental components of interaction, motivating us to ask whether current LLM agents can reliably perform them together (Riemer et al., 2025).

How can these capabilities be diagnosed and trained in a controlled way? Games provide a natural substrate because their information structure, utilities, legal actions, and strategic consequences can be specified and verified exactly. Recent work has accordingly used multi-agent games and self-play to train strategic or transferable reasoning in LLMs (Yuan et al., 2026; Liu et al., 2026; Lyu et al., 2026; He et al., 2026b). However, these environments are primarily designed around game-level performance rather than separate supervision of interactive-state inference and the planning that follows from it. Compact self-play games make scalable training possible but often abstract away the combination of latent information and heterogeneous incentives, whereas richer social games can entangle deduction, communication, deception, commitment and execution within the same trajectory (Light et al., 2023). To obtain precise supervision for the interaction structure above, we instead extend an existing commitment-based multi-party negotiation game (Benac et al., 2026), which makes the environment becomes incomplete information and mixed incentives. Incomplete information makes relevant properties of other agents latent and therefore creates an inference problem, while mixed incentives make those inferred properties consequential for planning because the value of an interaction depends on what the other agents know, prefer, or are likely to do. Their combination captures an important class of real interactions in which agents act on behalf of different principals—for example, coordinating around private scheduling constraints (Zou et al., 2026),

negotiating under asymmetric private objectives (O’Neill et al., 2026), or interacting repeatedly on behalf of distinct economic interests (Li et al., 2026b).

Using this controlled setting, we separately diagnose failures in interactive-state inference and interaction planning, then construct strategic game slices that provide oracle supervision for both. We train on these slices and evaluate whether the resulting improvements generalize to unseen strategic configurations and transfer to external multi-agent environments.

Our contributions are threefold:

- **A controlled diagnosis of interactive reasoning in LLM-based MAS.** Grounded in classical interactive decision making, we study interactive-state inference and interaction planning in incomplete-information, mixed-incentive coordination, and separately measure failures in the two while retaining their sequential dependence.
- **A compact game construction for verifiable social-reasoning supervision.** We show how private preferences, consequential commitments, and information-revealing actions can instantiate a practically relevant interactive-decision problem while providing exact latent-state targets and counterfactual action values.
- **Targeted game supervision and transfer.** We construct strategic game slices around diagnosed interaction decisions, use their oracle structure to supervise inference and planning, and test whether the resulting capabilities generalize to unseen strategic configurations and external multi-agent environments.

2 RELATED WORKS

Classical multi-agent decision making couples inference about other agents with downstream action selection, from opponent models and interactive beliefs (Carmel & Markovitch, 1996; Gmytrasiewicz & Doshi, 2005; Albrecht & Ramamoorthy, 2013) to belief-space coordination, inference from actions, and opponent-aware planning (Macindoe et al., 2012; Wu et al., 2021; Foerster et al., 2019; Huang et al., 2024). For LLM agents, explicit mental-state or behavioral representations support collaboration, adaptation, trust tracking, and planning (Li et al., 2023; Cross et al., 2025; Jha et al., 2026; Mu et al., 2026; Zhou et al., 2026; Feng et al., 2026; Goel et al., 2026; Hwang et al., 2026). However, accurate mental-state prediction need not yield useful behavior (Riemer et al., 2025; Sobotka et al., 2026).

Recent MAS benchmarks expose strategic failures under private information, mixed incentives, and imperfect communication. Commitment-based negotiation, self-negotiated contracts, and multi-party conquest model consequential agreements and coalition dynamics (Benac et al., 2026; Wyse et al., 2026; O’Neill et al., 2026); CalBench, SOTOPIA-TOM, and SNEAK make information disclosure and privacy strategically consequential (Zou et al., 2026; YS et al., 2026; Avsian & Heck, 2026). Broader evaluations cover collaboration, controlled interaction, long-horizon commerce, and communication barriers (Zhu et al., 2025; Chen et al., 2026; Li et al., 2026b; Xuan et al., 2026). Diagnostic work further finds failures in distributed information pooling, distributed computation, dynamic grounding, and belief-sensitive negotiation (Li et al., 2026a; Zhang et al., 2026b; Yao et al., 2026; Zhang et al., 2026a; Agashe et al., 2025); these complement broad failure taxonomies and accounts of long-lived peer interaction (Cemri et al., 2025; Anthropic, 2026).

Game-based training has been used to induce transferable strategic and social reasoning through self-play, multi-game reinforcement learning, and social reasoning supervision (Yuan et al., 2026; Liu et al., 2026; Lyu et al., 2026; He et al., 2026b; Hua et al., 2026; He et al., 2026a; Wu et al., 2026). Cross-environment cooperation and Other-Play study generalization across tasks and partners (Jha et al., 2025; Hu et al., 2020). Curriculum work selects terminal-proximal or high-learning-value subgames (West et al., 2020; Chen et al., 2024), while LLM curricula can also induce undesirable strategic abstractions (Madmoun & Lahlou, 2026). Similarity-based inference has also been proposed as a route to cooperation among foundation-model agents (Meulemans et al., 2026).

3 PRELIMINARIES

3.1 FROM INTERACTIVE UNCERTAINTY TO TWO CAPABILITIES

Coordination with private information is naturally an interactive partially observable decision problem. For agent i , an I-POMDP is $\mathcal{I}_i = \langle IS_i, A, T_i, \Omega_i, O_i, R_i \rangle$, $IS_i = S \times \prod_{j \neq i} M_j$ where S is the physical state, M_j models agent j , A is the joint action space, and T_i, O_i, R_i specify transition, observation, and reward (Gmytrasiewicz & Doshi, 2005). Uncertainty therefore covers not only the world but also others’ information, preferences, and policies. This view underlies opponent modelling and belief-space coordination (Carmel & Markovitch, 1996; Huang et al., 2024; Goel et al., 2026); the observation–belief–action links nevertheless remain fragile for LLM agents (Sobotka et al., 2026; Jha et al., 2026). Let $b_t^i \in \Delta(IS_i)$ be agent i ’s belief after public history h_t . An observation updates it as $b_{t+1}^i = \tau_i(b_t^i, a_t^i, o_{t+1}^i)$, while an action is selected by belief-conditioned planning, $a_t^{i,*} \in \arg \max_{a \in A_i} \mathbb{E} \left[\sum_{k=t}^T \gamma^{k-t} R_i(x_k, a_k) \mid b_t^i, a_t^i = a, \pi_{-i} \right]$. We call these computations *partner inference* (B), which maintains the part of b_t^i predictive of others, and *partner-conditioned planning* (P), which evaluates actions under that belief. They form the loop $B_t \rightarrow P_t \rightarrow a_t^i \rightarrow o_{t+1}^i \rightarrow B_{t+1}$: belief changes the best interaction, while the chosen interaction determines the next evidence. This is a functional decomposition, not a claim about separate neural modules.

3.2 BENAC-I: NEGOTIATION WITH PRIVATE PREFERENCES

To expose both computations while retaining exact supervision, we extend the commitment-based negotiation game BENAC (Benac et al., 2026). We call the resulting incomplete-information variant BENAC-I. An instance is

$$\mathcal{E} = \langle \mathcal{N}, \mathbf{K}, \mathcal{G}, \Gamma, \delta, \rho, C_0, \mathcal{L}, \{\Theta_i\}_{i \in \mathcal{N}}, \mu_0, \mathbf{q} \rangle. \quad (1)$$

Here $\mathcal{N}$ is the player set, K_i counts i ’s binary commitment coordinates, and each public goal $g \in \mathcal{G}$ has requirements $\Gamma_g \subseteq \{(i, k) : k < K_i\}$ and mode $\delta_g \in \{\text{binary}, \text{linear}\}$. The proposer schedule ρ and legal-action relation $\mathcal{L}$ govern play from the initially empty, irreversible commitment state C_0 . Each player privately observes a preference type $v_i \in \Theta_i \subseteq \{-1, 0, +1\}^{|\mathcal{G}|}$, drawn from common prior μ_0 ; entries mean AVOID, NEUTRAL, or WANT. Vector $\mathbf{q}$ records remaining investigation budgets. For commitment state C , goal satisfaction and terminal utility are

$$S_g(C) = \begin{cases} \mathbf{1}[c_{ik} = 1, \forall (i, k) \in \Gamma_g], & \delta_g = \text{binary}, \\ |\Gamma_g|^{-1} \sum_{(i, k) \in \Gamma_g} c_{ik}, & \delta_g = \text{linear}, \end{cases} \quad U_i(C; v_i) = \sum_g v_{ig} S_g(C). \quad (2)$$

On a proposal turn, a player may PASS, OFFER new commitments, or use INVESTIGATE(j, g) when $q_i > 0$. Acceptance irreversibly binds both sides’ proposed additions; rejection binds none; choosing from a two-offer proposal binds only the selected option. Investigation instead consumes the turn and one unit of q_i , leaves C unchanged, and privately reveals v_{jg} ; the target, but not the answer, is public. Negotiation therefore supplies indirect evidence and investigation supplies costly direct evidence within the same action space.

Reference teacher and labels. The teacher is an information-limited policy profile, not an omniscient player. At an information set $I_j = (h, v_j, r_j)$ —public history, own preferences, and private investigation results—it chooses

$$\text{supp } \pi_j^*(\cdot \mid I_j) \subseteq \arg \max_{a \in \mathcal{L}(I_j)} \mathbb{E}_{v_{-j} \sim b_j(\cdot \mid I_j)} [U_j(C_T; v_j) \mid a, \pi_{-j}^*], \quad (3)$$

breaking own-utility ties by total utility to others and mixing over residual ties. We enumerate the finite game tree, initialize legal actions uniformly, and synchronously update information-set best responses until the whole contingent profile is stable. We retain it only after a unilateral-deviation check; cycles and resource-limit failures yield no label. This defines a reproducible reference, not a unique or general-sum Nash equilibrium.

The same profile supplies both labels. An observed action gives the exact finite-support update $b_{t+1}^i(v_{-i}) \propto b_t^i(v_{-i}) \pi_j^*(u_t^j \mid I_t^j, v_j)$, treating i ’s actions as interventions and conditioning on any

private investigation result; this is the B target. For P , exact continuation gives $Q_i^*(I_i, a)$ for every legal action and hence every optimal action. Both labels thus share one declared behavioral model and terminal objective.

4 DIAGNOSE

We next test whether an LLM can recover the partner state needed for a decision (B), choose an effective interaction when that state is supplied (P), and compose the two. Final game reward alone cannot distinguish these failures: a poor action may follow either a wrong judgment or poor use of a correct one, while two errors may occasionally cancel.

4.1 MATCHED SEMANTIC INTERVENTIONS

We construct 16 held-out two-player structures, evenly divided between binary and linear goals. Each structure yields a matched pair. In the *voluntary* case, the displayed partner actions are observations generated by the fixed reference policy; in the *preset* case, the same actions are imposed interventions and hence provide no evidence about preference. The two cases end at exactly the same physical state and have the same legal actions, per-world payoff table, and fixed continuation policy. Changing evidential status therefore changes only the posterior weights used to evaluate this otherwise identical planning problem. Selection uses teacher quantities only. In every pair, the correct semantic judgment changes and the two exact reference-optimal action sets are disjoint, certifying that inference is decision-relevant rather than incidental.

The model-facing interface is the one used by our agent. For the single unknown partner–goal preference, B returns (i) the complete subset of WANT, NEUTRAL, and AVOID that remains possible and (ii) the uniquely most supported value, or UNDETERMINED. It never returns probabilities. We then query fresh planning contexts with either the model judgment $\hat{B}$ or the correct semantic judgment B^* . These controlled planning prompts retain the complete current decision state—the binding commitments, remaining turns, investigation quota, legal actions, and fixed continuation policy—but remove the behavioral chronology from which B was inferred. Before inserting the judgment, the voluntary and preset planning requests are byte-identical. Thus the supplied judgment is the only route by which their evidential distinction reaches P . A third condition asks for a native history-to-action decision without an explicit judgment. Exact posteriors and per-world payoffs remain in the backend and are used only for scoring. In an auxiliary condition on eight voluntary structures, B additionally receives a qualitative reference-planner statement—e.g., that the observed sequence is impossible, more likely, or less likely under each preference—but still receives no numerical belief. This tests whether failures arise from modeling the partner’s plan or from subsequently integrating that behavioral evidence.

Let $F_c(a, \theta)$ be the certified payoff of legal action a in case c when the unknown preference is θ , under the same fixed continuation, and let b_c^* be the backend posterior. We score every action using

$$V_c(a) = \sum_{\theta} b_c^*(\theta) F_c(a, \theta), \quad \text{Reg}_c(a) = \max_{a'} V_c(a') - V_c(a). \quad (4)$$

Thus an action prompted by an incorrect judgment is still evaluated in the correct world distribution. We also measure the paired judgment-repair effect

$$\Delta_B = V_c(\hat{a}(B^*)) - V_c(\hat{a}(\hat{B})), \quad (5)$$

using deterministic decoding so that the two calls differ only in the supplied judgment. A positive value means that repairing B improves this planner; it is not an additive estimate of a standalone “B error.” Invalid tool outputs are missing rather than replaced with a default action.

4.2 INFERENCE FAILURE AND CLEAN PLANNING INTERVENTION

Table 1 reports the completed inference-side results for Qwen3-4B-Instruct-2507. Exact B requires both the complete possible set and the correct favored field. The controlled planning and repair entries are left blank pending the history-free rerun; the earlier values are not reused because their planning prompts retained the evidence supplied to B .

Table 1: Native-semantic diagnosis. Inference means weight the 16 structures equally. Dashes mark the pending history-free controlled- P rerun.

Quantity	Result
Independent structures / matched cases	16 / 32
Same state, actions, and payoff table	16/16
Different B^* and disjoint optimal-action sets	16/16
Certified decision gap, minimum / mean	0.17 / 0.50
Native semantic B exact	9.4%
Possible-set exact / favored exact	43.8% / 12.5%
Mean P regret given B^*	—
Positive regret given B^*	—
Mean $\hat{B} \rightarrow P$ composition regret	—
Native end-to-end regret	1.11
Action changed under B repair	—
Mean paired judgment-repair effect Δ_B	—
Qualitative partner-plan assistance: B exact gain	0.0 pp

The task rows first establish that inference matters: changing only evidential status changes the optimum, with a nontrivial minimum decision gap. Yet the model recovers the full semantic B in only 9.4% of cases. The much higher possible-set accuracy than favored accuracy shows that retaining candidates is easier than comparing their behavioral support; errors are dominated by unsupported additions rather than exclusions. Qualitative partner-plan assistance yields no exact-accuracy gain on its eight cases, so supplying the likelihood ordering alone does not make the model integrate that evidence correctly.

The native history-to-action condition has mean regret 1.11, showing that the integrated decision problem remains difficult, but it cannot by itself localize the failure to inference or planning. The history-free controlled P calls are required for that attribution: once completed, regret under B^* will measure planning with the inference result fixed, while Δ_B will measure whether changing that result alters the same planner’s utility. These two quantities have different meanings and will not be subtracted as additive error components.

Illustrative failure patterns. One completed trace exposes a specific inference failure. In the preset case of `binary-12066`, the prompt says that Blair was required to add CEDAR and that preset events are non-evidential. The model nevertheless favors WANT for Blair’s Library preference because Cedar appears in Library’s requirements; the correct judgment is the full set with UNDETERMINED. The explanation therefore treats a required physical transition as preference evidence even when the prompt explicitly identifies it as an intervention. We select the planning-side trace only after the clean P rerun, so that it cannot be explained by the model rereading the original behavioral evidence.

5 EXPERIMENT

6 CONCLUSION

LIMITATION AND FUTURE WORK

AI USE STATEMENT

We used generative AI tools to assist with literature discovery, reference formatting, and drafting and editing portions of this manuscript. All AI-assisted text and bibliographic metadata were reviewed by the authors. The authors take responsibility for the final content of the paper, including its claims, experimental artifacts, and citations.

ETHICS STATEMENT

This work studies LLM-agent interaction in a synthetic, simulated negotiation environment and does not involve human participants or personal data. The environment includes mixed incentives

and strategic information disclosure, which can surface behaviors such as manipulation or unhelpful information sharing. We use these behaviors only for controlled evaluation and training of coordination capabilities. We document the game rules, oracle policies, and limitations so that the resulting methods are not interpreted as deployment policies for real-world autonomous agents.

REPRODUCIBILITY STATEMENT

The main text and appendix specify the BENAC-P game rules, information structure, oracle belief filtering, and reference planning procedure. The appendix reports the diagnostic positions, prompts, evaluated model version, decoding configuration, controls, aggregation procedure, random seed, and provenance information used for the reported results. These details are intended to make the diagnostic suite and its measurements reproducible.

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A NATIVE-SEMANTIC DIAGNOSTIC PROTOCOL

Frozen matched structures. The diagnostic contains 16 held-out structures, eight with binary goals and eight with linear goals. Each structure contributes a voluntary and a preset case. The two cases have byte-identical current physical states, legal-action sets, per-world payoff tables, and fixed continuation policies. They differ only in whether the displayed prefix was chosen by the reference partner or externally imposed. Model outputs were not used for selection. CPU certificates verify that the two cases have different correct semantic judgments and disjoint exact reference-optimal action sets. The gap between the best and best non-optimal action ranges from 0.17 to 1.00 terminal-utility units (mean 0.50), ruling out nominal differences caused only by numerical ties.

The queried uncertainty is one partner-goal preference in $\{\text{WANT}, \text{NEUTRAL}, \text{AVOID}\}$. The deployed B interface returns the complete supported subset and either its uniquely favored member or UNDETERMINED; it does not return probabilities. Both the model and correct judgments use identical serialization. The B prompt necessarily identifies whether an event was observed or imposed; the controlled P prompt contains neither that evidence nor its provenance. Condition names, case identifiers, correctness claims, posterior probabilities, and payoff values are absent from model prompts. Exact posteriors remain in the scorer and are used with the common per-world payoff table to compute Eq. 4.

Calls and interventions. For each case we obtain: (i) a native semantic B judgment; (ii) a fresh P call supplied with that judgment; (iii) a fresh P call supplied with B^* ; and (iv) a native history-to-action call with no explicit judgment. The two controlled P calls include the current binding state, remaining turns, investigation quota, known preferences, legal actions, and fixed partner policy, but exclude the behavioral chronology and its voluntary/preset provenance. Their tools, sampling parameters, and request seed are identical; only the serialized judgment may differ. Moreover, after removing that judgment, the controlled- P request is byte-identical across the two cases in each matched structure. If the model judgment is invalid, its composition cell is missing rather than replaced by a default action. On eight voluntary cases, a fifth call supplies B with the reference planner’s qualitative likelihood ordering while still withholding all numeric probabilities.

Selection and scoring use the fixed teacher only. In particular, changing a judgment never recomputes the partner continuation policy, and per-world payoffs respect the focal player’s information set rather than optimizing a different continuation separately in each hidden world. The paired quantity Δ_B is therefore the effect of changing the semantic input to the same model planner. It can have either sign and is not combined with planning regret as an additive error decomposition.

Execution and determinism. We evaluate Qwen3-4B-Instruct-2507 through vLLM and the Hermes tool-call parser, using greedy decoding ($T = 0$, $\text{top-}p = 1$, $\text{top-}k = -1$), a 4,096-token completion limit, and three replicas. The vLLM workers enable batch-invariant execution; repeated requests must agree even when production concurrency and chunked prefill are retained. We additionally require that whenever $\hat{B} = B^*$, the two controlled- P request bodies (including their seed) are identical and return the same action. Replicas are determinism checks, not independent statistical samples. The final clean- P run path and coverage counts will be inserted after completion.

Aggregation. We first average valid replicas within each case and the two matched cases within each structure, then weight the 16 structures equally. Completed B results correspond to 3/32 exact judgments, 14/32 exact possible sets, and 4/32 correct favored fields; qualitative partner-plan assistance converts none of its eight cases into an exact semantic judgment. Controlled- P coverage and paired repair counts will be filled from the history-free run, without treating the three deterministic replicas as additional evidence.

The frozen v6 suite stores the model-facing cases, teacher-only certificates, selection record, and both clean controlled- P and native end-to-end requests. CPU checks reject any controlled planning prompt containing a behavioral chronology or provenance marker and verify byte identity across each matched pair before judgment insertion. Hashes in the suite manifest detect changes to frozen cases and label dependencies before inference.